Below are solutions/hints on some of the questions. There is no guarantee of completeness or correctness; please watch out for typos.

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1. A particle on a one dimensional potential is subject to a potential that is constant for $x \notin [-a, a]$:

$$V(x) = \begin{cases} 0 & \text{for } x < -a \\ \text{unknown} & \text{for } -a < x < a \\ V_0 & \text{for } x > a \end{cases}$$

The Schroedinger equation for this system is found/known to have a solution with energy E with the following forms in the regions outside $[-a, a]$:

$$\psi(x) = \alpha e^{ik_1 x} + \beta e^{-ik_1 x} \qquad \text{for } x < -a$$
$$\psi(x) = \gamma e^{ik_2 x} \qquad \text{for } x > a$$

where

$$\alpha = \frac{k_1 + ik_2}{k_1 - ik_2} \qquad \beta = \frac{ik_1}{k_1 - ik_2} \qquad \gamma = \frac{\sqrt{k_1 k_2}}{k_1 - ik_2}$$

(a) **[4 pts.]** Express k_1 in terms of E.

Hint/Solution →
You can put in the wavefunction in the left ($x < a$) region into the Schroedinger equation with $V(x) = 0$. This will yield the following relation between E and k_1. However, you should actually already know the answer, since we worked this out a few times in class.

$$E = \frac{\hbar^2 1^2}{2m} \qquad k_1 = \sqrt{\frac{2mE}{\hbar^2}}$$

(b) **[4 pts.]** Is E larger or smaller than V_0? Explain why.

Hint/Solution →
$E > V_0$, because we have a plane wave solution in the region where $V(x) = V_0$. If $E < V_0$, we would have exponential solutions in this region.

(c) **[5 pts.]** Express k_2 in terms of E and V_0.

Hint/Solution →
Again, you can insert $\psi(x) = \gamma e^{ik_2 x}$ in the Schroedinger equation for $x > a$, and this will give the relation

$$E - V_0 \;=\; \frac{\hbar^2 k_2^2}{2m} \qquad k_2 \;=\; \sqrt{\frac{2m}{\hbar^2}(E - V_0)}$$

but you should in fact already know the answer.

(d) **[3 pts.]** If we want to interpret this as a scattering problem, which parts of the wavefunction would you interpret as incident wave, reflected wave, and transmitted wave?

Hint/Solution →

incident	$\alpha e^{ik_1 x}$
reflected	$\beta e^{-ik_1 x}$
transmitted	$\gamma e^{ik_2 x}$

(e) **[4 pts.]** Calculate the reflection coefficient.

Hint/Solution →

$$R \;=\; \left|\frac{\beta}{\alpha}\right|^2 \;=\; \left|\frac{ik_1}{k_1 + ik_2}\right|^2 \;=\; \frac{k_1^2}{k_1^2 + k_2^2}$$

Note there are no momentum factors. This is explained in the following.

(f) [**5 pts.**] Calculate the transmission coefficient.
(Careful: remember momemntum factors.)

Hint/Solution →
The momentum factors appear because the transmission coefficient is the ratio of transmitted current to incident current, and the current density carried by a plane wave $Ae^{ipx/\hbar}$ is $|A|^2 p/m$. (Please do prove this to yourself using the definition of the current density.) Thus the transmission coefficient is

$$T = \frac{\text{transmitted current}}{\text{incident current}} = \frac{|\gamma|^2 \hbar k_2/m}{|\alpha|^2 \hbar k_1/m} = \left|\frac{\gamma}{\alpha}\right|^2 \times \frac{k_2}{k_1}$$

Note that, in the reflection coefficient, the momentum factors cancel (both are $\hbar k_1$) and hence are not needed.

$$T = \left|\frac{\gamma}{\alpha}\right|^2 \times \frac{k_2}{k_1} = \left|\frac{\sqrt{k_1 k_2}}{k_1 + ik_2}\right|^2 \frac{k_2}{k_1} = \frac{k_2^2}{k_1^2 + k_2^2}$$

(g) [**3 pts.**] Check that the sum of the two coefficients makes sense.

Hint/Solution →

$$R + T = \frac{k_1^2}{k_1^2 + k_2^2} + \frac{k_2^2}{k_1^2 + k_2^2} = 1$$

Interpreting the incident plane wave as a steady current/stream of particles, it's obvious that the particles either get reflected or they get transmitted. The sum of the reflected and transmitted currents should be equal to the incident current: otherwise particles are being created or destroyed. This means that the sum of reflection and transmission coefficients needs to be equal to unity.

2. Suppose $\psi_0(x)$ is a properly-normalized wavefunction with

$$\langle \hat{x} \rangle_{\psi_0} = \langle \psi_0 | \hat{x} | \psi_0 \rangle = x_0 \qquad \langle \hat{p} \rangle_{\psi_0} = \langle \psi_0 | \hat{p} | \psi_0 \rangle = p_0$$

where x_0 and p_0 are constants. The subscript indicates the state in which the expectation values are being calculated. Define a new wavefunction

$$\psi_{new}(x) = e^{iqx/\hbar}\, \psi_0(x)$$

where q is a real quantity having dimensions of momentum.

(a) **[6 pts.]** What is the expectation value $\langle \hat{x} \rangle_{\psi_{new}}$ in the state $\psi_{new}(x)$?

Hint/Solution →

$$\langle \hat{x} \rangle_{\psi_{new}} = \langle \psi_{new} | \hat{x} | \psi_{new} \rangle = \int_{-\infty}^{\infty} x \left| \psi_{new}(x) \right|^2 dx$$

$$= \int_{-\infty}^{\infty} x \left| e^{iqx/\hbar} \psi_0(x) \right|^2 dx = \int_{-\infty}^{\infty} x \left| e^{iqx/\hbar} \right|^2 \left| \psi_0(x) \right|^2 dx$$

$$= \int_{-\infty}^{\infty} x \left| \psi_0(x) \right|^2 dx = \langle \hat{x} \rangle_{\psi_0} = x_0$$

(b) **[7 pts.]** What is the expectation value $\langle \hat{p} \rangle_{\psi_{new}}$ in the state $\psi_{new}(x)$?

Hint/Solution →

$$\langle \hat{p} \rangle_{\psi_{new}} = \langle \psi_{new} | \hat{p} | \psi_{new} \rangle = \int_{-\infty}^{\infty} \psi_{new}(x)^* \hat{p} \psi_{new}(x) dx$$

$$= -i\hbar \int_{-\infty}^{\infty} \psi_{new}(x)^* \frac{d}{dx} \psi_{new}(x) dx$$

Now

$$\frac{d}{dx}\psi_{new}(x) = \frac{d}{dx}\left[e^{iqx/\hbar}\, \psi_0(x) \right] = \frac{iq}{\hbar} e^{iqx/\hbar}\, \psi_0(x) + e^{iqx/\hbar} \frac{d}{dx}\psi_0(x)$$

so that

$$\psi_{new}(x)^* \frac{d}{dx}\psi_{new}(x)$$

$$= \left[e^{-iqx/\hbar}\, \psi_0(x)^* \right] \left(\frac{iq}{\hbar} e^{iqx/\hbar}\, \psi_0(x) + e^{iqx/\hbar} \frac{d}{dx}\psi_0(x) \right)$$

$$= \frac{iq}{\hbar} \left| \psi_0(x) \right|^2 + \psi_0(x)^* \frac{d}{dx}\psi_0(x)$$

Thus

$$\langle \hat{p} \rangle_{\psi_{new}} = -i\hbar \frac{iq}{\hbar} \int_{-\infty}^{\infty} |\psi_0(x)|^2 \, dx \; - \; i\hbar \int_{-\infty}^{\infty} \psi_0(x)^* \frac{d}{dx} \psi_0(x) dx$$

$$= q \; + \; \int_{-\infty}^{\infty} \psi_0(x)^* \hat{p} \psi_0(x) dx \; = \; q \; + \; \langle \hat{p} \rangle_{\psi_0} \; = \; q \; + \; p_0$$

A ‘COMMON’ MISTAKE $\longrightarrow$

Here’s a WRONG calculation:

$$|\psi_{new}\rangle \; = \; e^{iqx/\hbar} \; |\psi_0\rangle$$

$$\Longrightarrow \; \langle \hat{p} \rangle_{\psi_{new}} \; = \; \langle \psi_{new} | \hat{p} | \psi_{new} \rangle$$

$$= \; e^{-iqx/\hbar} \langle \psi_0 | \hat{p} | \psi_0 \rangle e^{iqx/\hbar} \qquad \text{WRONG}$$

$$= \; \langle \psi_0 | \hat{p} | \psi_0 \rangle \; = \; p_0 \qquad \text{WRONG}$$

Where did we go wrong? We mixed the abstract bra-ket notation with the specific real-space representation by writing expressions mixing $e^{iqx/\hbar}$ (specific real-space expression) and bra’s and ket’s (in which the real-space dependence is hidden away). Clearly a bad idea. While the first line above is maybe not necessarily incorrect by itself, it is susceptible to mistreatment due to the mixing of notation.

(c) **[2 pts.]** Based on your results, interpret in one sentence the physical significance of multiplying a wavefunction by the factor $e^{iqx/\hbar}$.

Hint/Solution $\rightarrow$

Multiplying by this phase factor $e^{iqx/\hbar}$ keeps the average of the position unchanged, and shifts the momentum by the quantity q.

Multiplying by such a phase factor is sometimes informally called ‘boosting’ the wavefunction.

3. Consider a particle with mass m in the state described by wavefunction $\psi(x)$, which we write in amplitude-phase form:

$$\psi(x) \;=\; A(x)e^{i\theta(x)}$$

(a) **[SELF]** Write down the definition of the probability current density.

Hint/Solution →

$$j(x) \;=\; \frac{\hbar}{m}\mathrm{Im}\left[\psi(x)^*\frac{\partial}{\partial x}\psi(x)\right]$$

or

$$j(x) \;=\; \frac{i\hbar}{2m}\left[\psi(x)\frac{\partial}{\partial x}\psi(x)^* - \psi(x)^*\frac{\partial}{\partial x}\psi(x)\right]$$

The first form is usually easier to calculate with, but of course both should give the same answer. I use the first form below, but students should try out both ways to get the same result.

(b) **[5 pts.]** Show that the probability current density is given by

$$j(x) \;=\; [A(x)]^2\,\frac{\hbar}{m}\,\partial_x\theta(x)$$

i.e., the probability current is proportional to the *gradient* of the phase.

Hint/Solution →

$$\partial_x\psi \;=\; \partial_x\left[Ae^{i\theta}\right] \;=\; (\partial_x A)\,e^{i\theta} \;+\; A\partial_x e^{i\theta}$$
$$=\; (\partial_x A)\,e^{i\theta} \;+\; Ae^{i\theta}i\partial_x\theta$$

Multiplying by $\psi(x)^* = Ae^{-i\theta}$,

$$\psi^*\partial_x\psi \;=\; A\,(\partial_x A) \;+\; iA^2\partial_x\theta$$

so that

$$j \;=\; \frac{\hbar}{m}\mathrm{Im}\left[A\,(\partial_x A) \;+\; iA^2\partial_x\theta\right] \;=\; \frac{\hbar}{m}A^2\partial_x\theta \;=\; \frac{\hbar}{m}\,[A(x)]^2\,\partial_x\theta(x)$$

(c) **[2 pts.]** Show that there is no current in a region where the wavfunction is real.

Hint/Solution $\rightarrow$
In a region where the function is real, $\theta(x) = 0$, so that $\partial_x\theta(x) = 0$, which implies from the above result:

$$j = 0$$

Alternatively, using the longer definition $j = \frac{i\hbar}{2m}(\psi\partial_x\psi^* - \psi^*\partial_x\psi)$, we find that when the wavefunction is real ($\psi = \psi^*$), the current becomes

$$j \;=\; \frac{i\hbar}{2m}\left(\psi\partial_x\psi - \psi\partial_x\psi\right) \;=\; 0$$